

Datasets Analyzing Tornado Research

Gloria Le

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Datasets

- All files have 6 channels. -> Likely represent the same 6 variables across all tornado cases.
- Spatial resolutions vary a lot.
- Value ranges are mostly consistent. -> min ≈ -0.01 to -0.2 , max ≈ 1.60
- The max value (~ 1.60) is repeated -> Suspicious but interesting
- NaN percentages are generally low. (4% - 11%)



TOR5_BEST_BEFORE_win20d.tif

TOR5_BEST_AFTER_win50d.tif

- Size = 402x639 grid
- 6 channels
- Data is float32
- Data is highly positively correlated between channels
- Small dynamic range (-0.003705, 0.6036325) and (-0.002, 0.55031)

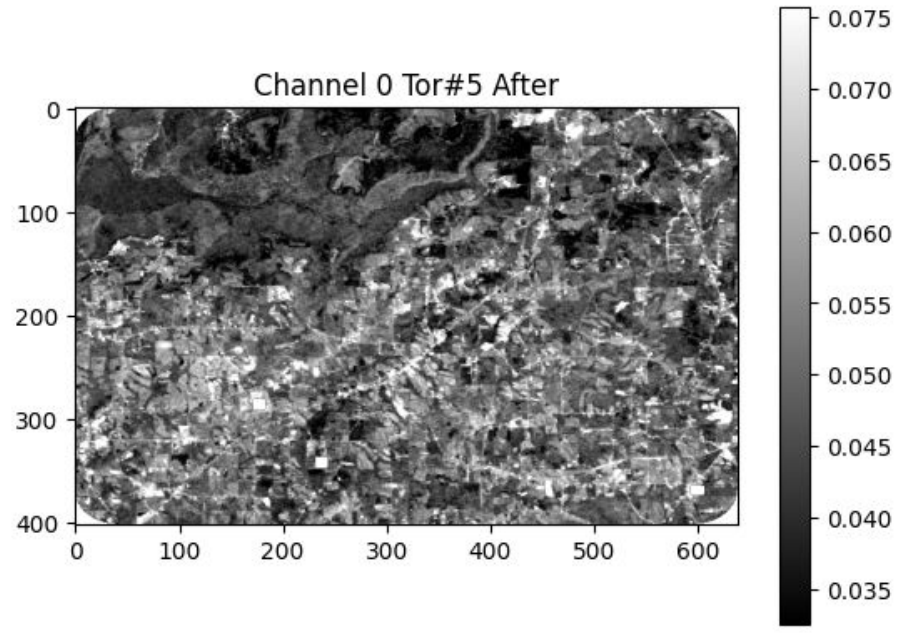
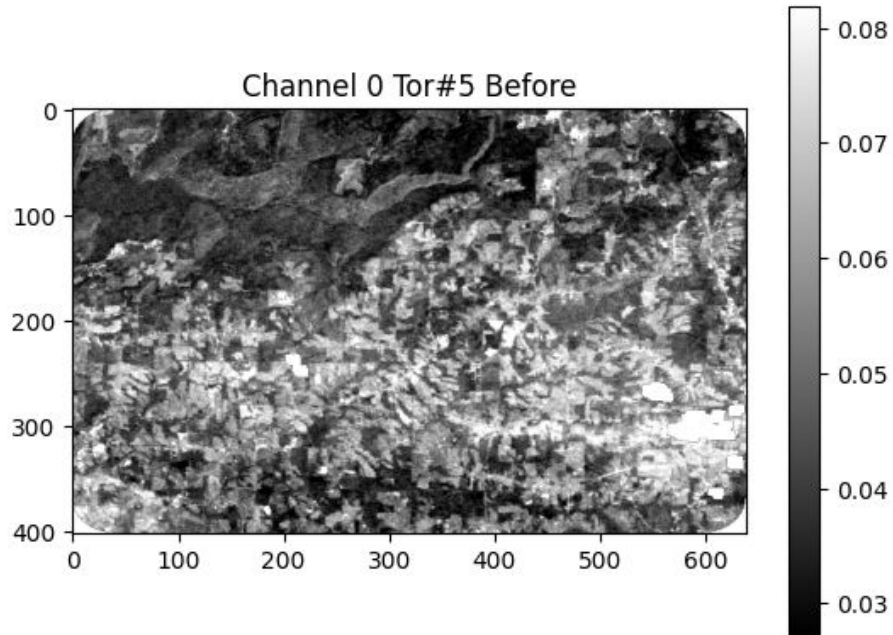
TOR5_BEST_BEFORE_win20d

```
[[1.      0.93226546 0.92181465 0.63171117 0.776887  0.82080988]
 [0.93226546 1.      0.95198049 0.74108545 0.81609155 0.84411882]
 [0.92181465 0.95198049 1.      0.69496233 0.88993545 0.91973895]
 [0.63171117 0.74108545 0.69496233 1.      0.75488587 0.67958201]
 [0.776887  0.81609155 0.88993545 0.75488587 1.      0.97547217]
 [0.82080988 0.84411882 0.91973895 0.67958201 0.97547217 1.      ]]
```

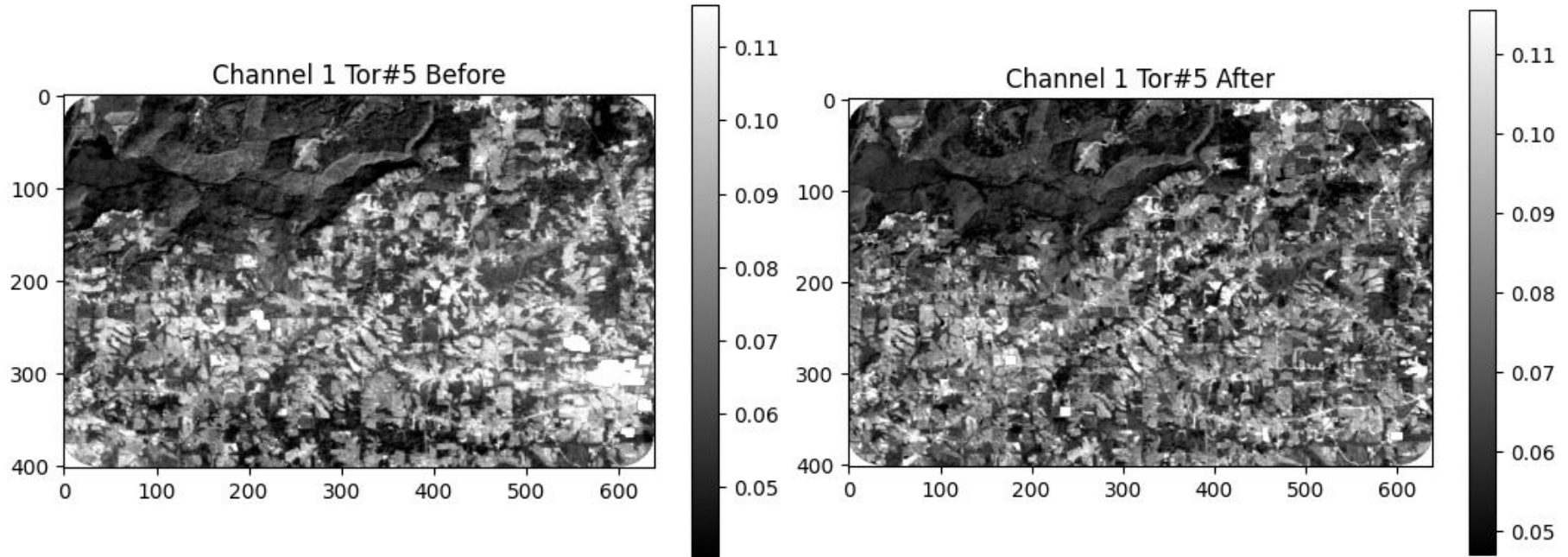
TOR5_BEST_AFTER_win50d

```
[[1.      0.92691196 0.92408727 0.43586314 0.704782  0.75918613]
 [0.92691196 1.      0.90804997 0.60968721 0.68969264 0.69284562]
 [0.92408727 0.90804997 1.      0.38471281 0.80681353 0.86754683]
 [0.43586314 0.60968721 0.38471281 1.      0.50323584 0.29262491]
 [0.704782  0.68969264 0.80681353 0.50323584 1.      0.94369522]
 [0.75918613 0.69284562 0.86754683 0.29262491 0.94369522 1.      ]]
```

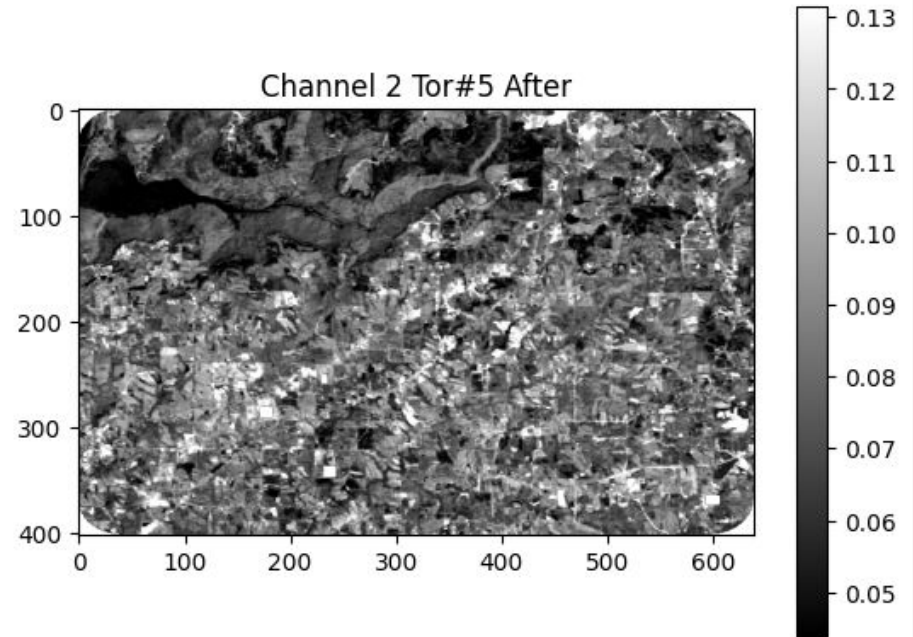
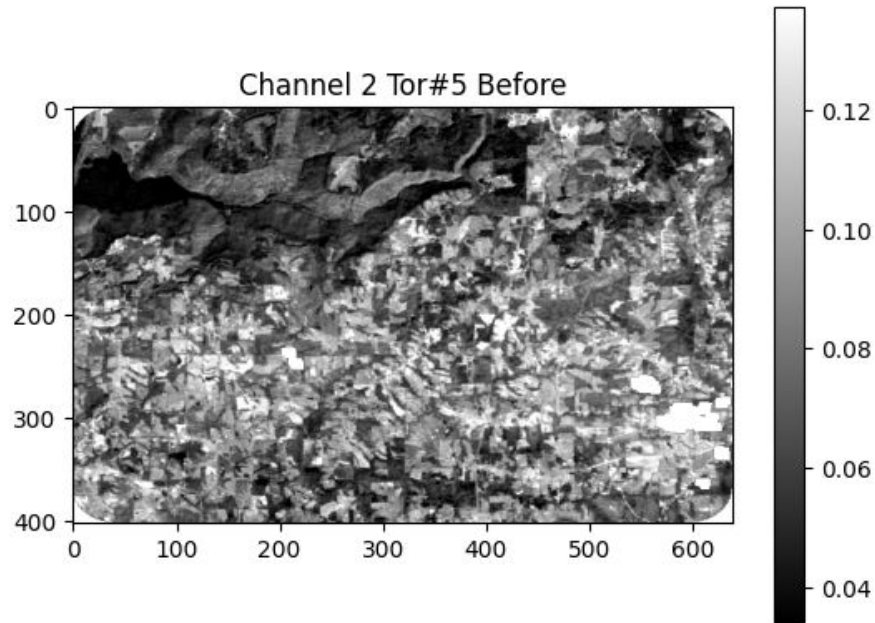
Tor#5 Before vs After



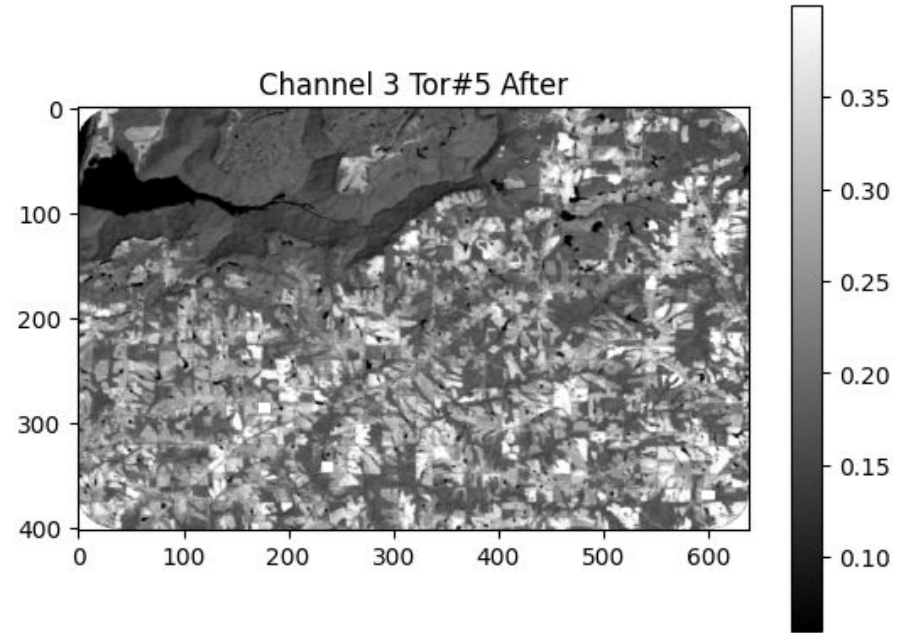
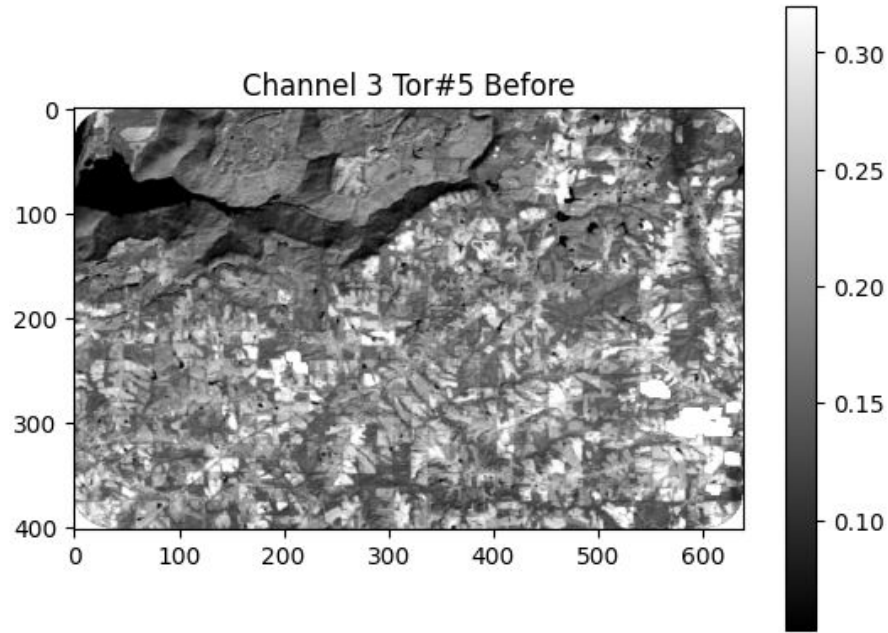
Tor#5 Before vs After



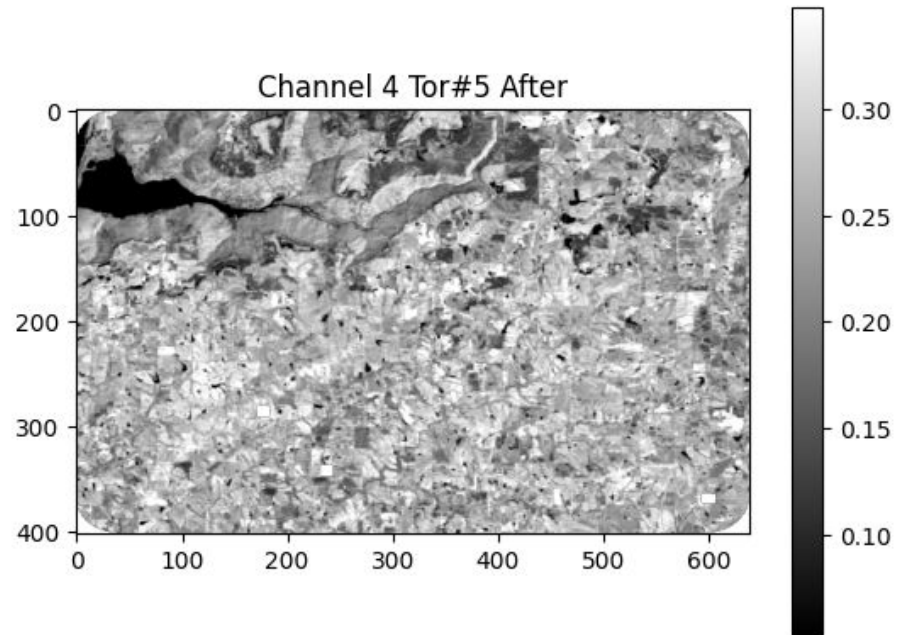
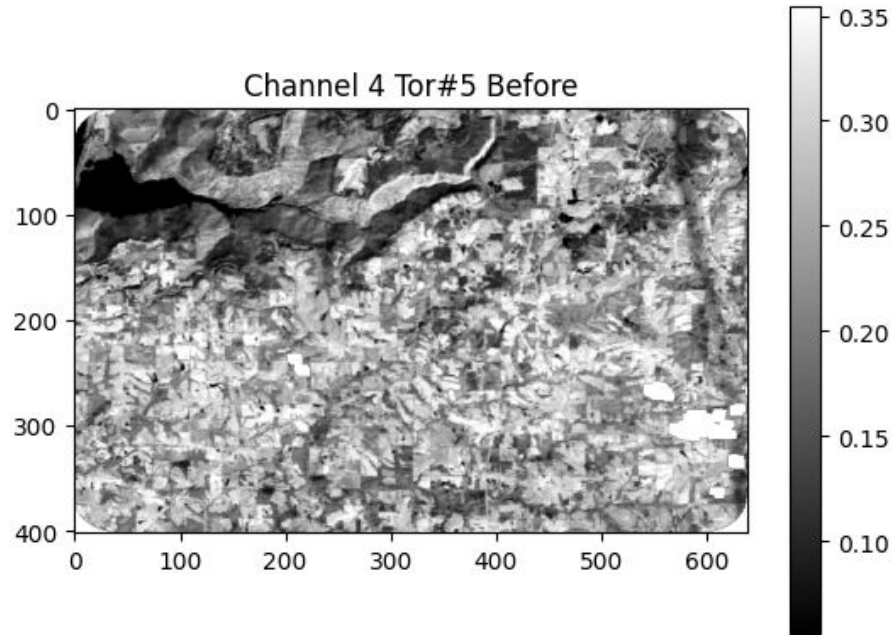
Tor#5 Before vs After



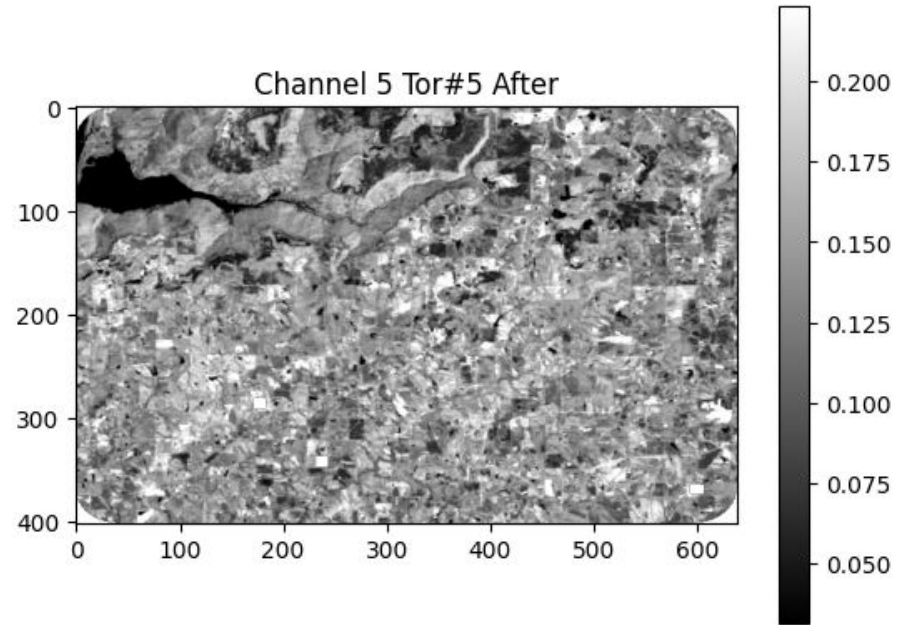
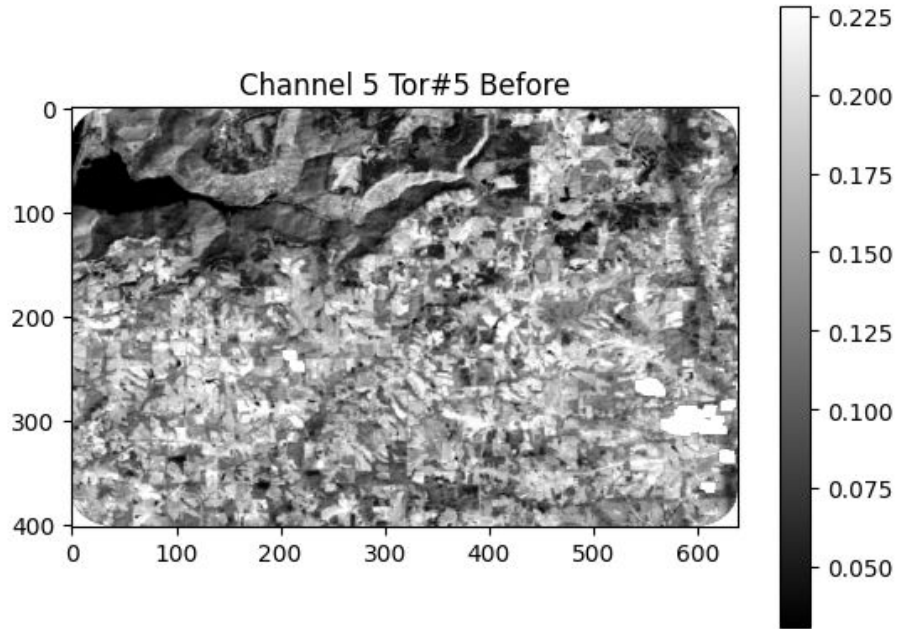
Tor#5 Before vs After



Tor#5 Before vs After

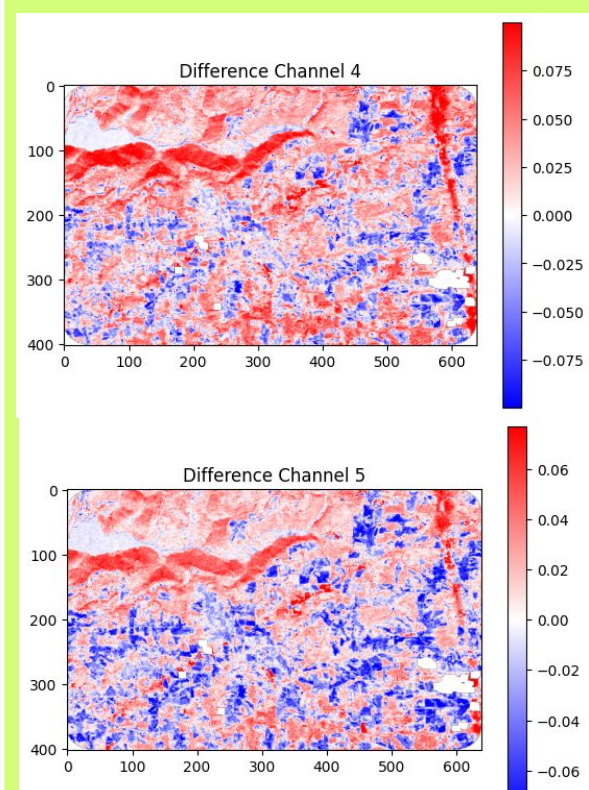
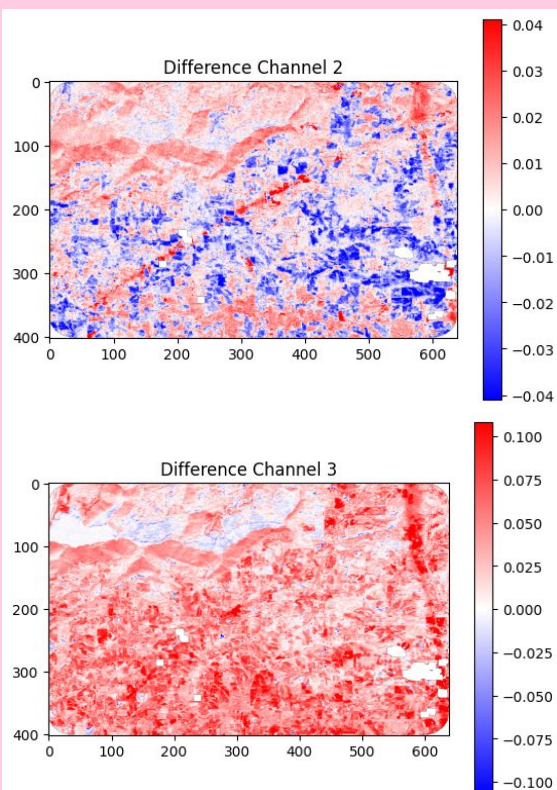
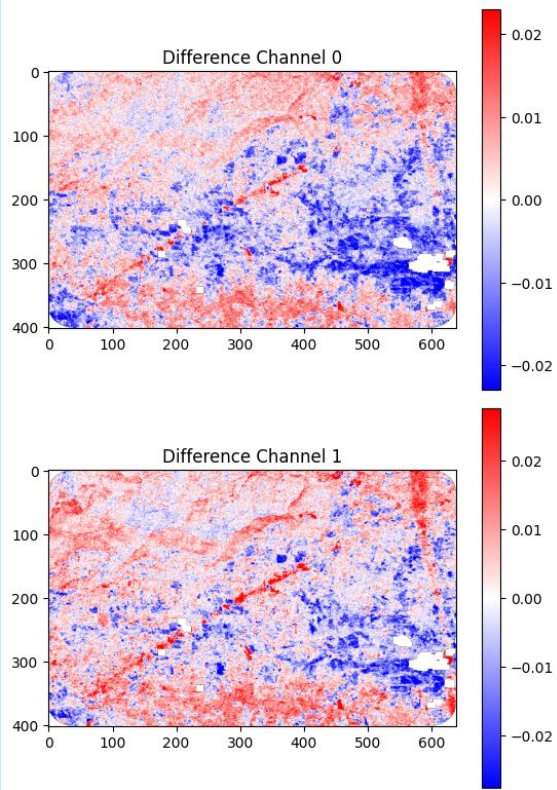


Tor#5 Before vs After



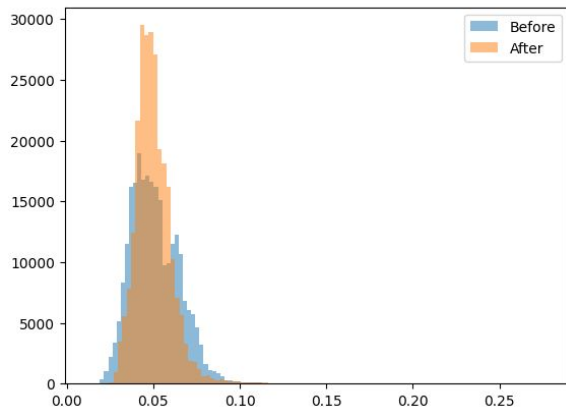
Difference between Before and After Tor#5

- Red = increased after
- Blue = decreased after

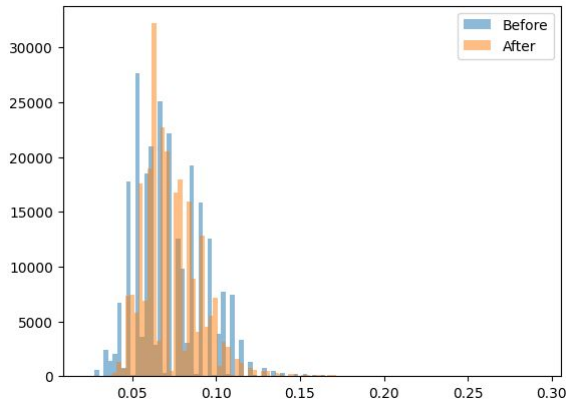


Distribution Shifts Tor#5

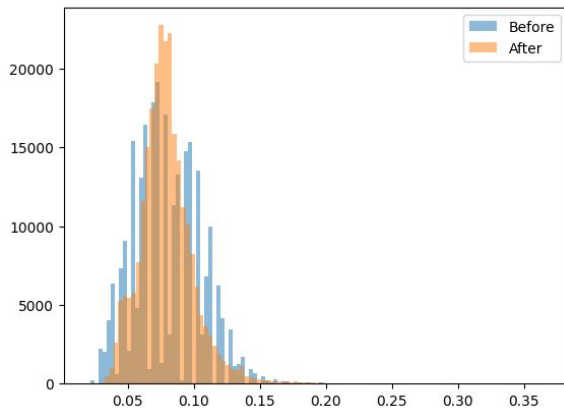
Channel 0 Distribution



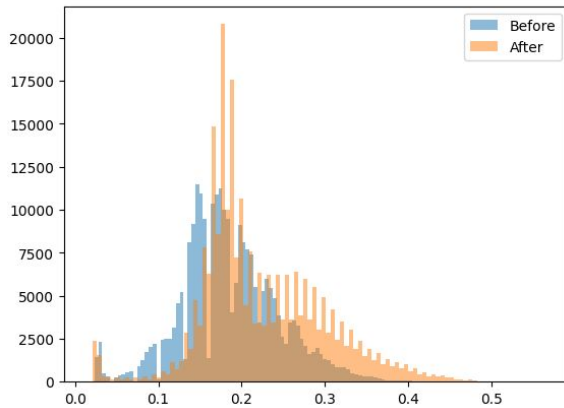
Channel 1 Distribution



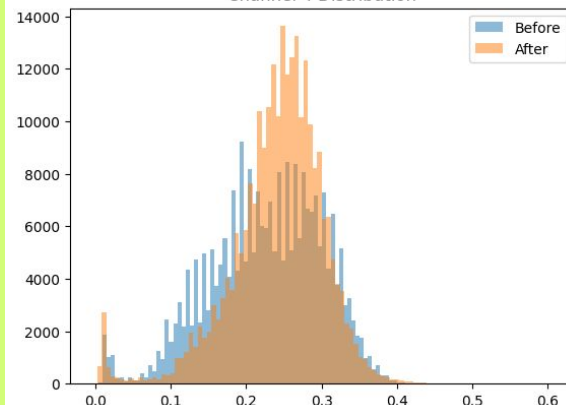
Channel 2 Distribution



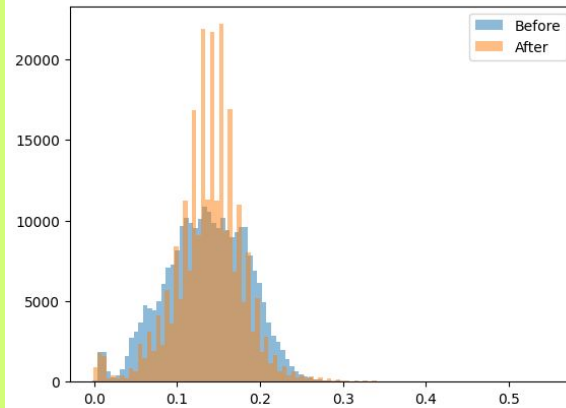
Channel 3 Distribution



Channel 4 Distribution



Channel 5 Distribution



Interpretation from Difference and Distribution Shift plots Tor#5

After consistently shows:

- Higher variance
- Higher extremes
- Systematic mean increase

-> Small but consistent statistical deviations.

-> Tornado event likely changes the environmental field in measurable ways.

```
Channel 0:
mean_before = 0.051094   mean_after = 0.050223902
variance_before = 0.00020452341   variance_after = 0.00011706973

Channel 1:
mean_before = 0.07210457   mean_after = 0.073556624
variance_before = 0.00042643325   variance_after = 0.00032166758

Channel 2:
mean_before = 0.07971262   mean_after = 0.079591356
variance_before = 0.0006864973   variance_after = 0.00046383915

Channel 3:
mean_before = 0.1861135   mean_after = 0.22334963
variance_before = 0.0035283542   variance_after = 0.0055659073

Channel 4:
mean_before = 0.22687523   mean_after = 0.24173257
variance_before = 0.0055332286   variance_after = 0.0038591677

Channel 5:
mean_before = 0.13717853   mean_after = 0.13826396
variance_before = 0.0024334162   variance_after = 0.0017557992
```



TOR62_BEST_BEFORE_win35d.tif

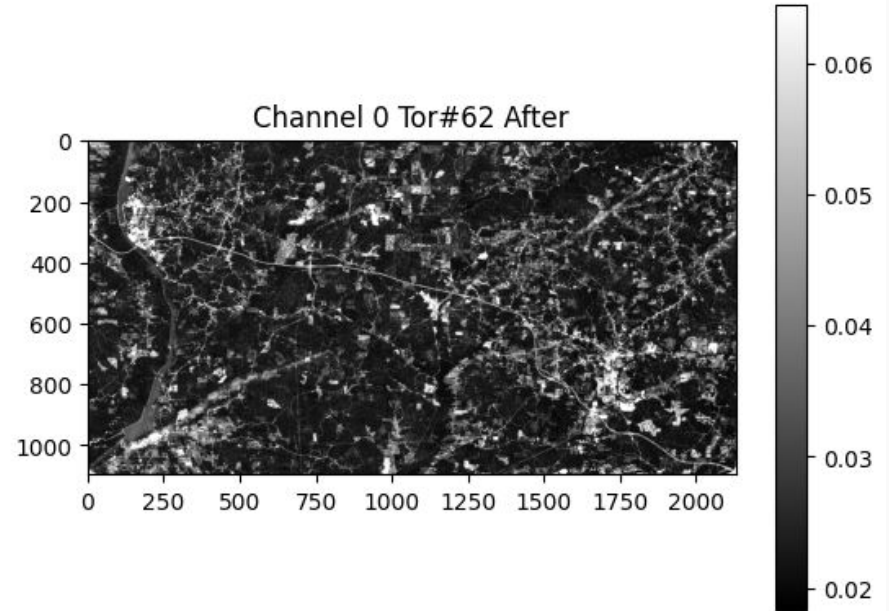
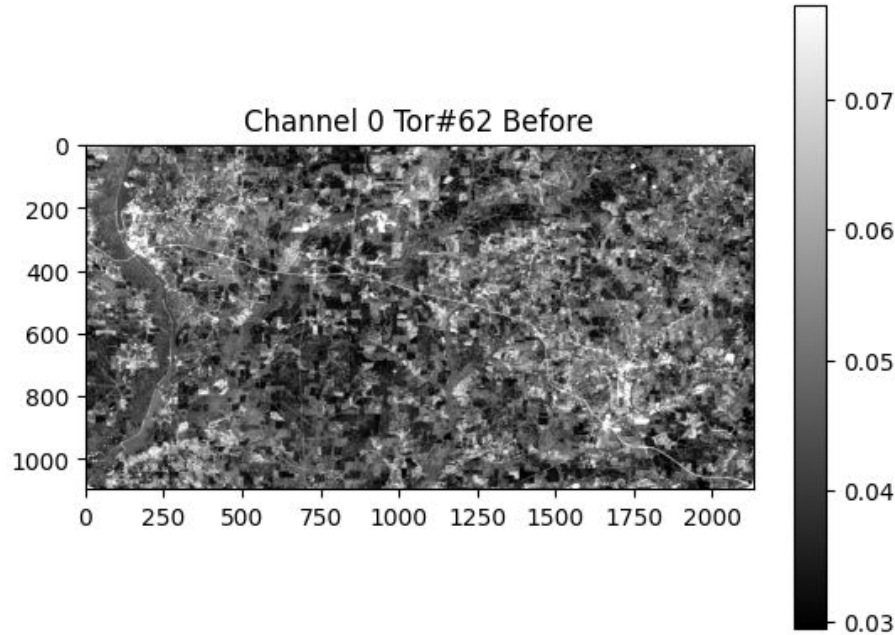
TOR62_BEST_AFTER_win35d.tif

- Size = 1096x2132 grid
- 6 channels
- Data is float32
- Bigger dynamic range (-0.010305, 1.6022125) and (-0.010415 1.6022125)
- Some very high positive correlations, also some moderate negative correlations.
 - > The channels are not all moving together anymore.
 - > Some features increase while others decrease spatially.

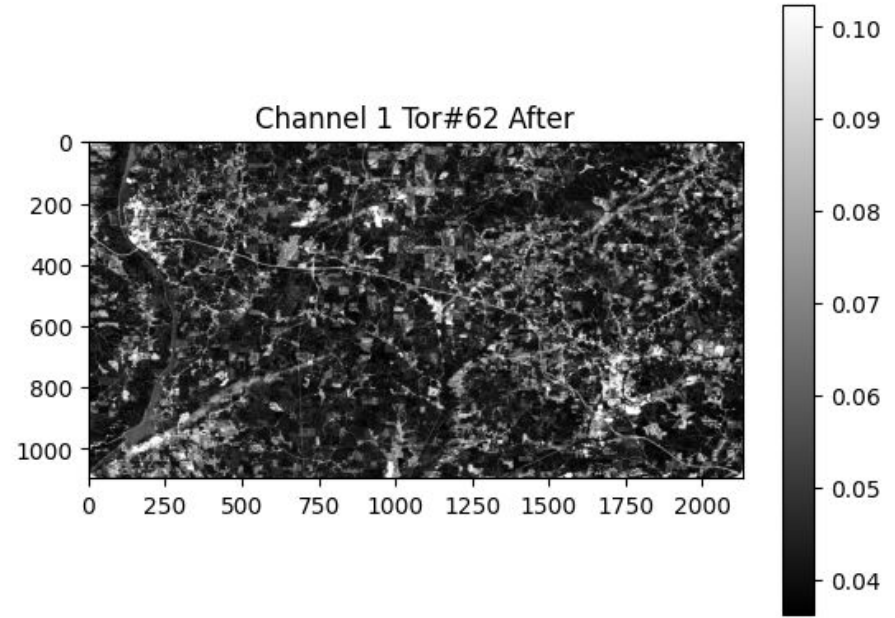
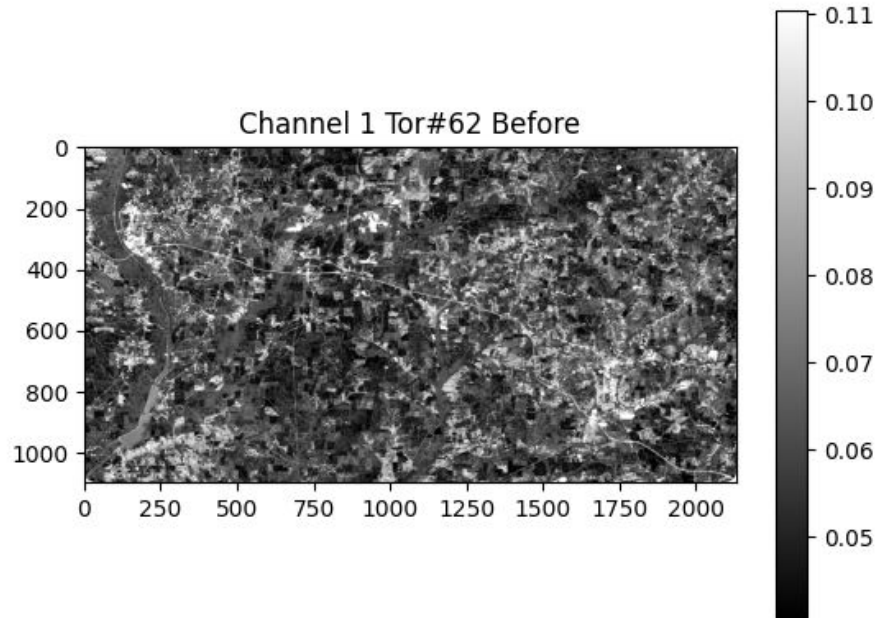
```
TOR62_BEST_BEFORE_win35d
[[1.          0.923162   0.91703714 0.33548852 0.77217323 0.82690335]
 [0.923162   1.          0.94679118 0.46511134 0.78303626 0.82933006]
 [0.91703714 0.94679118 1.          0.36137186 0.87142317 0.91861381]
 [0.33548852 0.46511134 0.36137186 1.          0.45818173 0.35943024]
 [0.77217323 0.78303626 0.87142317 0.45818173 1.          0.97135908]
 [0.82690335 0.82933006 0.91861381 0.35943024 0.97135908 1.          ]]

TOR62_BEST_AFTER_win35d
[[ 1.          0.89385964  0.88971398 -0.26276506  0.7042147  0.80183466]
 [ 0.89385964  1.          0.96102805 -0.18961005  0.82467723  0.89184225]
 [ 0.88971398  0.96102805  1.          -0.32794062  0.82015806  0.92298415]
 [-0.26276506 -0.18961005 -0.32794062  1.          0.04832492 -0.18579294]
 [ 0.7042147  0.82467723  0.82015806  0.04832492  1.          0.9492977 ]
 [ 0.80183466  0.89184225  0.92298415 -0.18579294  0.9492977  1.          ]]
```

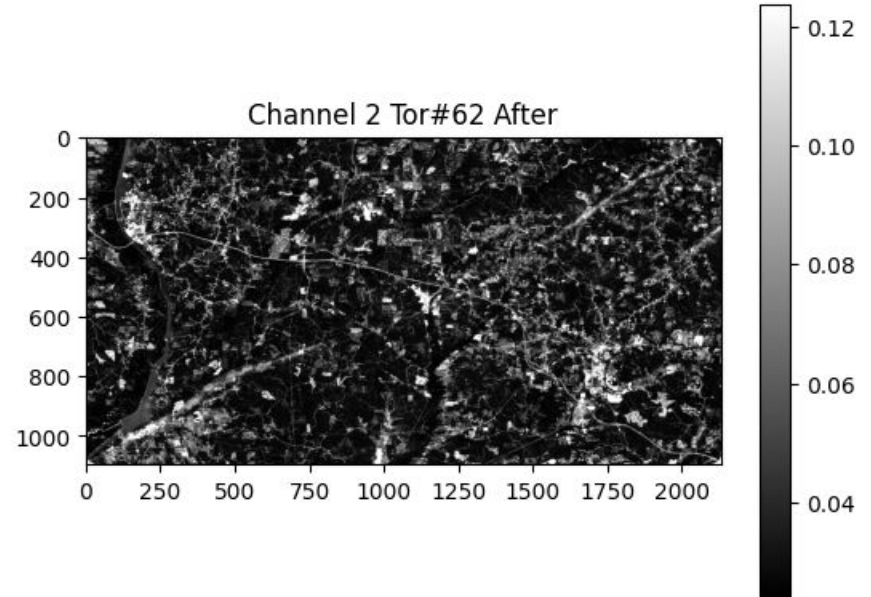
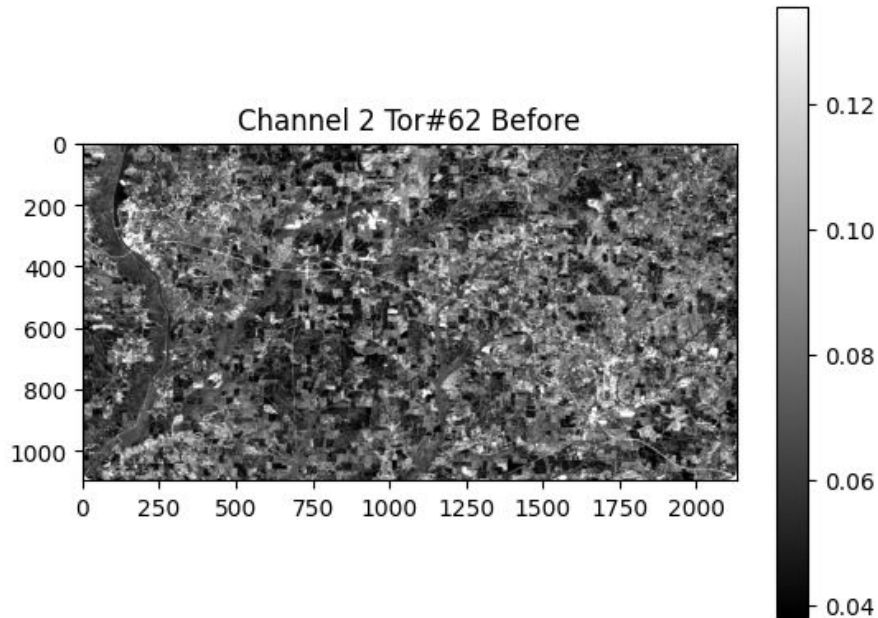
Tor#62 Before vs After



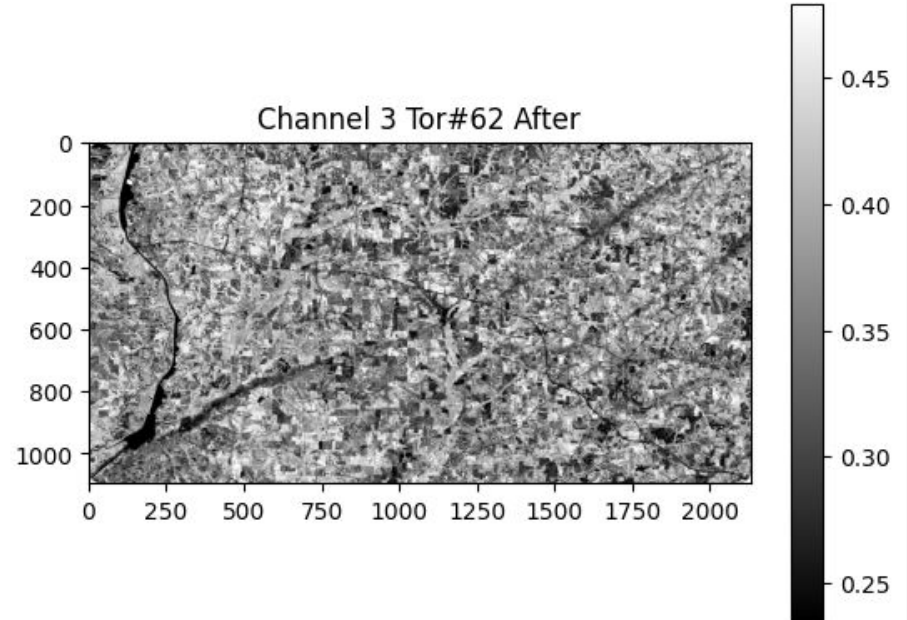
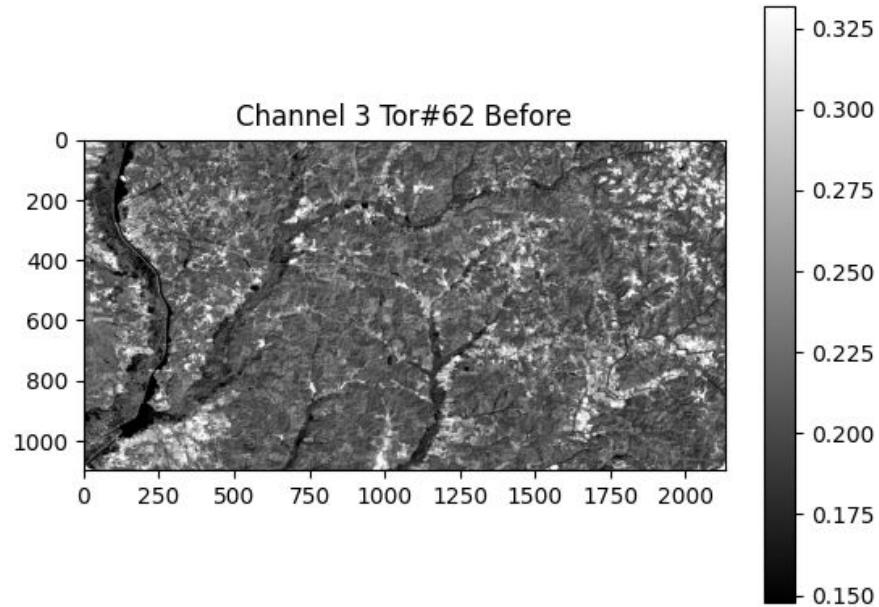
Tor#62 Before vs After



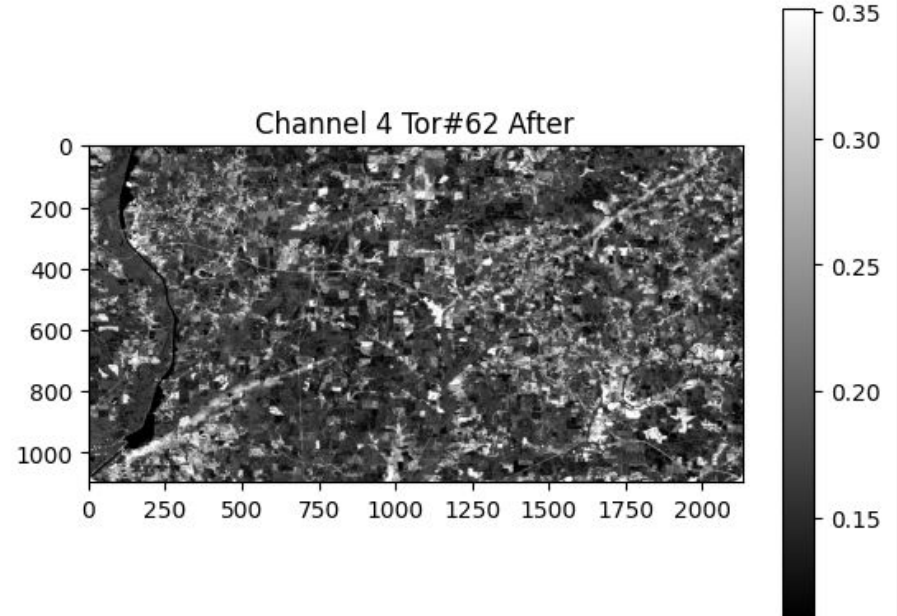
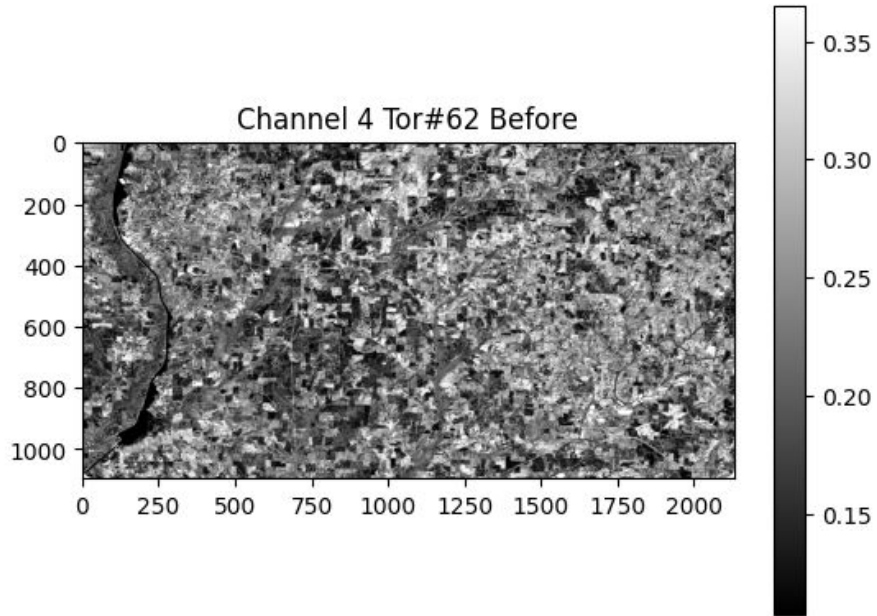
Tor#62 Before vs After



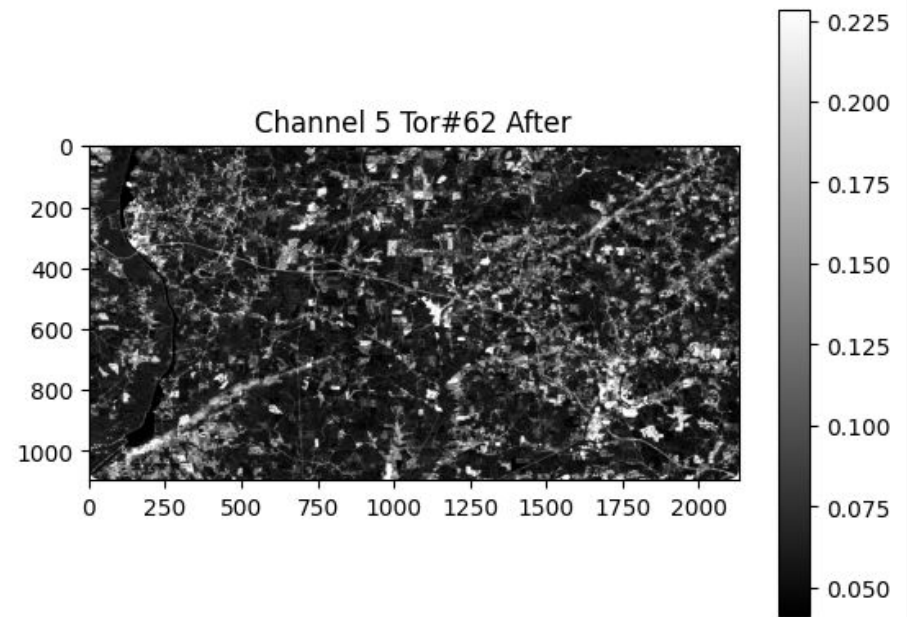
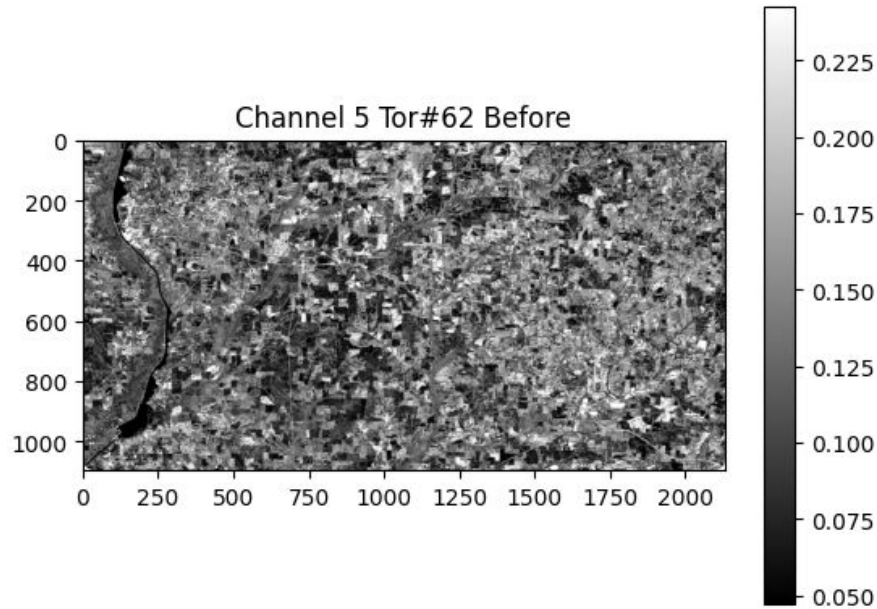
Tor#62 Before vs After



Tor#62 Before vs After

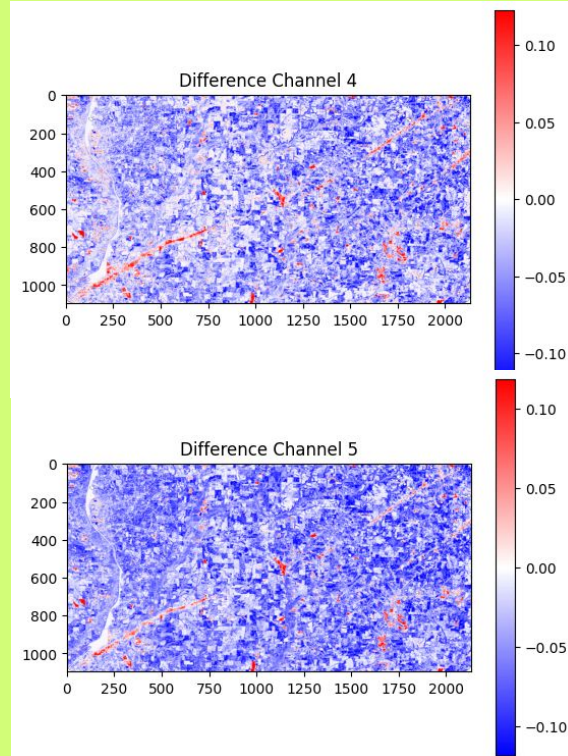
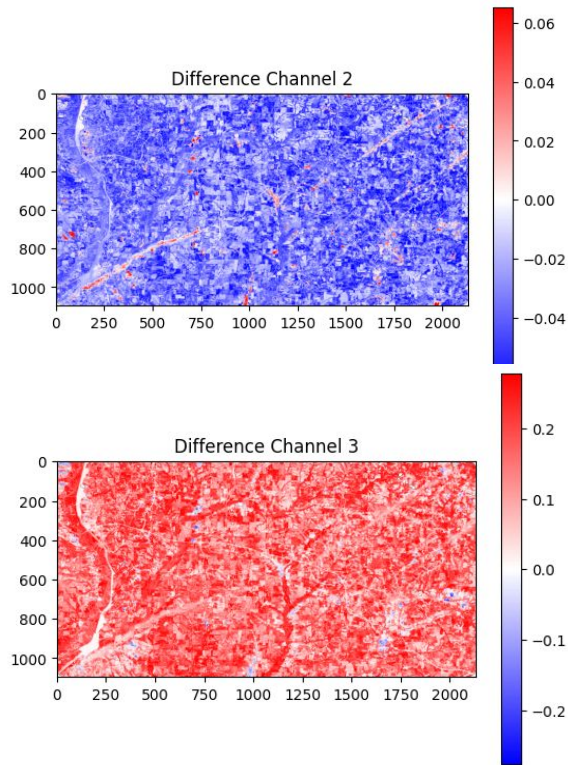
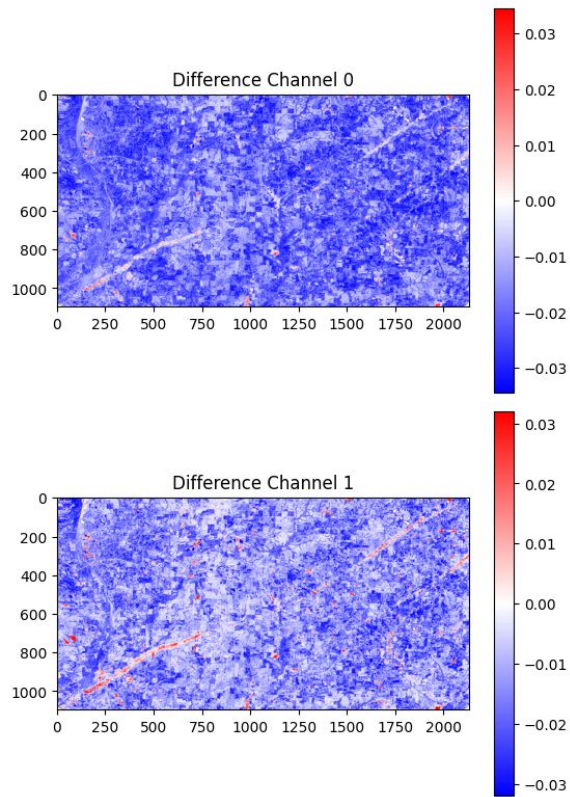


Tor#62 Before vs After

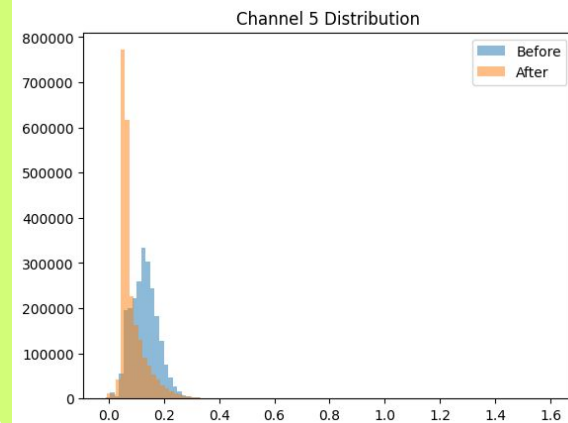
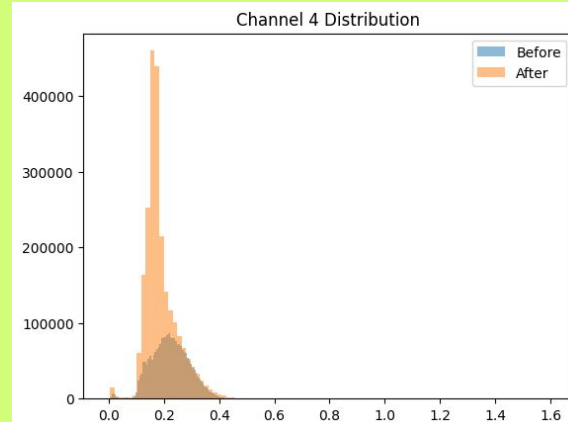
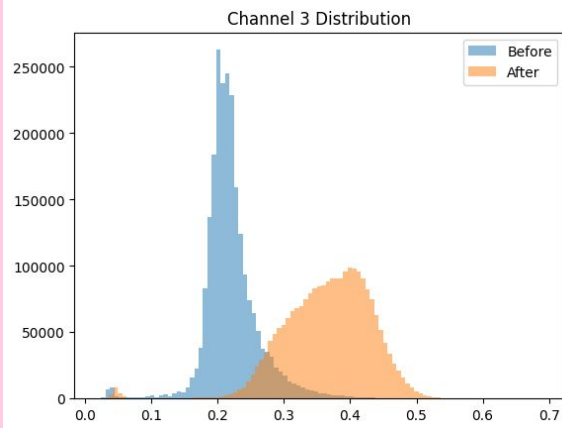
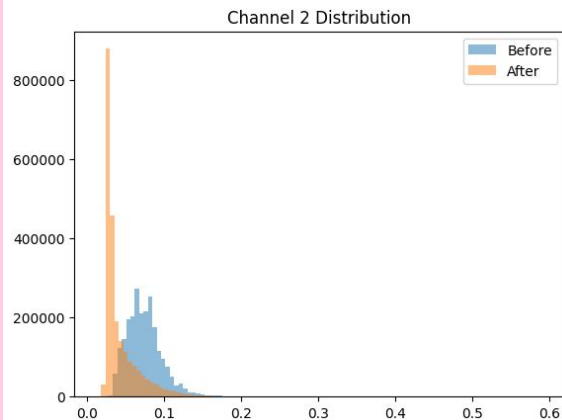
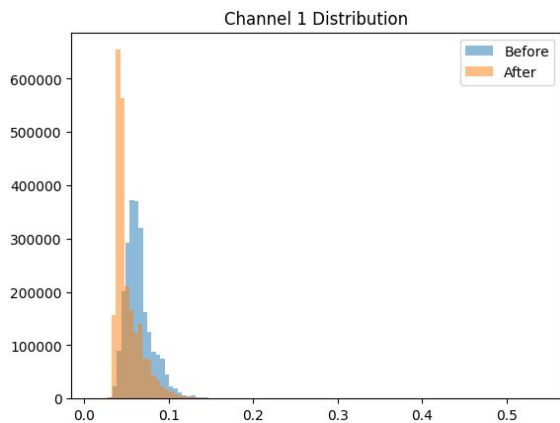
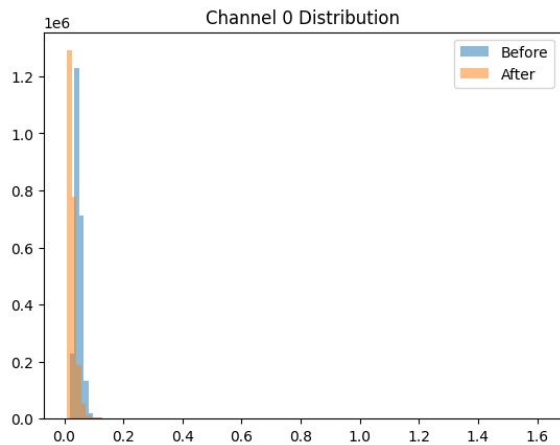


Difference between Before and After Tor#62

- Red = increased after
- Blue = decreased after



Distribution Shifts Tor#62



Interpretation from Difference and Distribution Shift plots Tor#62

After shows:

- Decrease in means and increase in variance
 - Not a uniform shift event
 - One specific feature spikes strongly
 - Channel 3 increases while other decreases. Indeed, correlation matrix shows negative correlation.
- > Might be the most informative feature.
- > TOR#62 is richer than TOR#5 as its features are interacting in more complex ways.

```
Channel 0:
mean_before = 0.047894876   mean_after = 0.02902824
variance_before = 0.00016574304   variance_after = 0.00018595728

Channel 1:
mean_before = 0.06514696   mean_after = 0.052355085
variance_before = 0.00033407836   variance_after = 0.00033983606

Channel 2:
mean_before = 0.074532665   mean_after = 0.04306544
variance_before = 0.0006321102   variance_after = 0.0007137194

Channel 3:
mean_before = 0.2203636   mean_after = 0.36577013
variance_before = 0.0018051947   variance_after = 0.0044462895

Channel 4:
mean_before = 0.22392735   mean_after = 0.18936573
variance_before = 0.004612502   variance_after = 0.0035564404

Channel 5:
mean_before = 0.13065928   mean_after = 0.08429411
variance_before = 0.0024993578   variance_after = 0.0022124876
```

Thank you!

Contact
le000415@umn.edu